# HARSHA S. BHAT

**Director of Research at CNRS**
**Teaching Prof. Ecole Polytechnique**
*Laboratoire de Géologie*
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## EDUCATION

| *Institution* | Degree | Field | Year |
|---|---|---|---|
| *École Normale Supérieure, France* | H.D.R.† | Supershear Earthquakes | 2021/01 |
| *Harvard University, USA* | Ph.D.* | Mechanical Sciences | 2007/06 |
| *Harvard University, USA* | M.S. | Engineering Sciences | 2002/06 |
| *NITK, India* | B.E. | Civil Engineering | 2001/06 |

*† Habilitation à Diriger des Recherches — * Supervised by Prof. J. R. Rice & Dr. R. Dmowska*

## CURRENT POSITION

| *Institution* | Period | Role |
|---|---|---|
| *École Normale Supérieure, France* | 2024/10 ▸ Present | CNRS Director of Research |
| *École Polytechnique, France* | 2022/09 ▸ Present | Teaching Professor (PCC) |

## PAST POSITIONS

| *Institution* | Period | Role |
|---|---|---|
| *École Normale Supérieure, France* | 2016/05 ▸ 2024/09 | CNRS Research Scientist |
| *NISER, India* | 2021/11 ▸ 2023/11 | Visiting Professor |
| *Institut de Physique du Globe de Paris, France* | 2012/01 ▸ 2016/05 | CNRS Research Scientist |
| *University of Southern California, USA* | 2010/03 ▸ 2011/12 | Asst. Professor (Research) |
| *University of Southern California, USA* | 2007/11 ▸ 2010/03 | Post Doctoral Fellow |
| *California Institute of Technology, USA* | 2007/11 ▸ 2010/03 | Visitor in Aeronautics |
| *Harvard University, USA* | 2007/05 ▸ 2007/10 | Post Doctoral Fellow |
| *Harvard University, USA* | 2001/11 ▸ 2007/05 | Grad. Research Associate |

## FUNDING & GRANTS†

- 2021–2025 ▸ 2M€ ERC Consolidator Grant, PERSISMO (Grant No. 865411)
- 2023–2027 ▸ 100k€ ANR SMEC
- 2023–2026 ▸ 100k€ ANR Univ. Tokyo SESAME
- 2023–2025 ▸ 66k€ SPARC Grant India
- 2018 ▸ 25k€ ENS Actions Incitatives
- 2017 ▸ 6k€ TelluS INSU — action ALEAS

## HONORS AND AWARDS

- 2018 CNRS Award for Doctoral Supervision and Research
- 2018 Grand Prix Michel Gouilloud Schlumberger, French Academy of Sciences
- 2006 Harvard University Certificate of Distinction in Teaching
- 2004 Harvard University Certificate of Distinction in Teaching
- 2003 Harvard University Certificate of Distinction in Teaching

## CURRENT TEACHING RESPONSIBILITIES AT ÉCOLE POLYTECHNIQUE†

Mécanique du Milieu continus · Mécanique des Solides · Mécanique de la Rupture · Mécanique des matériaux et des structures

## STUDENTS

| *Postdoctoral Associates* | | **Current Position** |
|---|---|---|
| *Ankit Gupta (India)* | 2024–2026 | |
| *Suli Yao (China)* | 2025–2027 | |
| *Navid Kheirdast (Iran)* | 2022–2025 | Post Doctoral Fellow at ISTEP |
| *Michelle Almakari (France)* | 2021–2023 | |
| *Carlos D. Villafuerte (Mexico)* | 2021–2023 | Asst. Prof. UNAM |
| *Ekeabino Momoh (Nigeria)* | 2019–2022 | AXA Postdoc Fellow |
| *Lucile Bruhat (France)* | 2018–2021 | Natural Catastrophe Risk Analyst at AXA |
| *Lisa Gordeliy (Russia)* | 2019–2019 | Senior Engineer ResFrac |
| *Marion Y. Thomas (France)* | 2014–2016 | CNRS Scientist at Université de Rennes |

| *PhD Students* | | **Current Position** |
|---|---|---|
| *Bharath Shanmugasundaram (India)* | 2024–2027 | |
| *Yishuo Zhou (China)* | 2024–2027 | |
| *Thomas Melkior (France)* | 2023–2026 | |
| *Caiyuan Fan (China)* | 2023–2027 | |
| *Jinhui Cheng (China)* | 2021–2024 | Postdoc Caltech |
| *Augustin Thomas (France)* | 2020–2023 | Research Scientist, BRGM |
| *Joseph Flores Cuba (Peru)* | 2020–2023 | Data Scientist PowerBI |
| *Claudia Hulbert (France)* | 2018–2021 | CEO Geolabe |
| *Samson Marty (France)* | 2017–2020 | Postdoc at Caltech |
| *Marshall A. Martinez (USA)* | 2014–2019 | Engineer at Joby Aviation |
| *Kurama Okubo (Japan)* | 2015–2018 | Research Scientist at NIED, Japan |
| *Pierre Romanet (France)* | 2014–2017 | Research Scientist at Cerema, France |
| *Vahe Gabuchian (USA)* | 2010–2015 | Research Scientist at Caltech |
| *François X. Passelègue (France)* | 2011–2014 | CNRS Scientist at GeoAzur, Nice |
| *Jonathan Mihaly (USA)* | 2008–2013 | Jet Propulsion Laboratory |
| *Michael Mello (USA)* | 2007–2012 | Teaching Professor at Caltech |

***Undergraduate and Masters Interns***

Valentin Marnat (2022) · Roxane Ferry (2021) · Jinhui Cheng (2020) · Phillipe Danre (2019) · Roxane Ferry (2019) · Hugo Lestrelin (2019) · Nicolas Mercury (2018) · Luc Illien (2018) · Phillipe Danre (2017) · Eleni Kolokytha (2015) · Victor Barolle (2015) · Kurama Okubo (2014) · Thibaut Perol (2013) · Lucile Bruhat (2012) · Marion Olives (2004) · Sonia Fliss (2003)

## PAST TEACHING ACTIVITIES†

Mécanique des Milieux Continus · Active Faults: Geometry · Seismic Ruptures and Scaling Laws · Introduction to Rock Physics · Mathematical Methods in the Sciences · Environmental Risks and Disasters · Ordinary and Partial Differential Equations · Complex and Fourier Analysis · Computational Solid and Structural Mechanics · Solid Mechanics · Introduction to the Mechanics of Solids · Mechanics of Fracture · Advanced Geomechanics · Mécanique de la Fracturation

*† Classes taught with various colleagues at Harvard, Caltech, IPGP, ENS*

## ORGANIZATION OF SCIENTIFIC MEETINGS

- **Jan 2026:** Earthquake Source: Mechanics, Seismology and Geology Workshop — NISER, India
- **Apr 2024:** Across the time scales, from earthquakes to earthquake cycle — EGU 2024
- **Apr 2023:** Across the time scales, from earthquakes to earthquake cycle — EGU 2023
- **Jun 2019:** Coupled Processes In Fracture Propagation In Geo-Materials: From Hydraulic Fractures To Earthquakes — CISM Advanced School, Udine, Italy
- **Apr 2015:** Seismological Society of America, Multiscale Modeling and Characterization of Fragmentation and Damage Patterns in Fault Zones
- **Dec 2014:** American Geophysical Union, Fault Zone Properties And Processes During Dynamic Ruptures

## INSTITUTIONAL RESPONSIBILITIES

- 2018–2024: Team Leader of Faults & Earthquakes Group, ENS
- 2018–2019: Co-organizer of the Internal Seminar, ENS
- 2025: Member HCERES committee for ISterre, Univ. Grenoble
- 2025 onwards: Member AGU Jason Morgan Award Jury

## REVIEWING ACTIVITIES

American Geophysical Union · Seismological Society of America · International Journal of Fracture · Geological Society of America · Science · Nature · Journal of the Mechanics and Physics of Solids · European Journal of Mechanics — A/Solids · Earth and Planetary Science Letters · Geophysical Research Letters · Journal of Structural Geology · Proceedings of the National Academies of Science, USA · Geology · Geophysical Journal International · Journal of Applied Mechanics · National Science Foundation · European Research Council · Nature Communications · Nature Geoscience · Science Advances

## BOOKS

Thomas, M. Y., Mitchell, T. M., and **Bhat,** H. S. (2017). ***Fault zone dynamic processes : Evolution of fault properties during seismic rupture, geophysical monograph 227***. American Geophysical Union (AGU) DOI: 10.1002/9781119156895.

Bizzarri, A., and **Bhat,** H. S. (2012). ***The mechanics of faulting: From laboratory to earthquakes***. Research Signpost.

## BOOK CHAPTERS

Thomas, M. Y., and **Bhat,** H. S. (2022b). ***Loi de friction et modélisation numérique du cycle sismique***. in *Le cycle sismique*, eds. Rolandone, F., ed. ISTE-Wiley. DOI: 10.51926/iste.9038.ch4.

Thomas, M. Y., and **Bhat,** H. S. (2022a). ***Friction laws and numerical modeling of the seismic cycle***. in *The seismic cycle: From observation to modeling*, eds. Rolandone, F., ed. ISTE-Wiley. DOI: 10.1002/9781394173709.ch4.

## PUBLICATIONS

Google Scholar: ZHskR34AAAAJ · ORCID: 0000-0003-0361-1854

Kheirdast, N., **Bhat,** H. S., Almakari, M., Gupta, A., Villafuerte, C., Thomas, M. Y., and Dubernet, P. (2026). ***Energy budget of spectrum of slip dynamics emerging from simplified model of fault and damage zone architecture***. to be subm. J. Geophys. Res.

Garagash, D. I., Brantut, N., **Bhat,** H. S., Schubnel, A., and Jolivet, R. (2026). ***Low effective stress along faults caused by upwelling fluid flow in laboratory and nature***. to be subm. J. Geophys. Res.

Gupta, A., **Bhat,** H. S., Faulkner, D. R., Bhattacharya, P., and Bollinger, L. (2026). ***A model for intraplate fault zones using rate and state friction law***. to be subm. J. Geophys. Res.

Cheng, J., **Bhat,** H. S., Almakari, M., Lecampion, B., and Dubernet, P. (2026). ***Quantifying the role of 3D fault geometry complexities on slow and fast earthquakes***. under review Geophys. Res. Lett DOI: 10.48550/arxiv.2602.16403.

Zhou, Y., Gupta, A., Aochi, H., Schubnel, A., Ide, S., and **Bhat,** H. S. (2026). ***Theoretical constraints on tidal triggering of slow earthquakes***. under review J. Geophys. Res DOI: 10.48550/arxiv.2602.06703.

Wang, C., Wang, P., Wu, B., **Bhat,** H. S., Bhattacharya, P., Xie, Y., Xia, K., and Schubnel, A. (2026). ***Complex fault slip behavior modulated by injection rate with implications to hazard mitigation of induced seismicity***. under review Science.

Almakari, M., Kheirdast, N., Villafuerte, C. D., Thomas, M. Y., Dubernet, P., Cheng, J., Gupta, A., Romanet, P., Chaillat, S., and **Bhat,** H. S. (2026). ***Fault volume digital twin to reproduce the full slip spectrum, scaling and statistical laws***. under review J. Geophys. Res DOI: 10.48550/arxiv.2509.04909.

Melkior, T., **Bhat,** H. S., and Amlani, F. (2025). ***Tsunami modeling with dynamic seafloors: A high-order solver validated with shallow water benchmarks***. under review J. Comput. Phys., submitted. DOI: 10.48550/2508.20596.

Michel, S., Scotti, O., Hok, S., **Bhat,** H. S., Kheirdast, N., Romanet, P., Almakari, M., and Cheng, J. (2026). ***A rate-and-state friction based criterion for the probability of earthquake fault jumps***. under review J. Geophys. Res DOI: 10.48550/arxiv.2501.15948.

Latour, S., Lebihain, M., **Bhat,** H. S., Twardzik, C., Bletery, Q., Hudnut, K. W., and Passelègue, F. (2025). ***Direct estimation of earthquake source properties from a single CCTV camera***. Science, DOI: 10.1126/science.adz1705.

Cheng, J., Almakari, M., Peruzzo, C., Lecampion, B., and **Bhat,** H. S. (2025). ***FASTDASH, a quasi-dynamic 3D seismic cycle model by using boundary element method with h-matrices***. Geophys. J. Int DOI: 10.1093/gji/ggaf230.

Momoh, E., **Bhat,** H. S., Tait, S., and Gerbault, M. (2025). ***Volumetric (dilatant) plasticity in geodynamic models and implications on thermal dissipation and strain localization***. Geophys. J. Int., DOI: 10.1093/gji/ggae463.

Ferry, R., Thomas, M. Y., **Bhat,** H. S., and Dubernet, P. (2025). ***Depth dependence of coseismic off-fault damage and its effects on rupture dynamics***. J. Geophys. Res., e2024JB029787. DOI: 10.1029/2024jb029787.

Petit, L., Olive, J.-A., Schubnel, A., Le Pourhiet, L., and **Bhat,** H. S. (2024). ***A brittle constitutive law for long-term tectonic modeling based on sub-critical crack growth***. to appear in Geochem. Geophys. Geosyst., DOI: 10.1029/2023gc011229.

Jeandet-Ribes, L., Thomas, M. Y., and **Bhat,** H. S. (2023). ***On the importance of setting 3-d stress field in simulations of on- and off-fault deformation***. Geophys. J. Int., DOI: 10.1093/gji/ggad401.

Marty, S., Schubnel, A., **Bhat,** H. S., Aubry, J., Fukuyama, E., Latour, S., Nielsen, S., and Madariaga, R. (2023). ***Nucleation of laboratory earthquakes: Quantitative analysis and scalings***. J. Geophys. Res., 128. DOI: 10.1029/2022jb026294.

Amlani, F., **Bhat,** H. S., Simons, W. J. F., Schubnel, A., Vigny, C., Rosakis, A. J., Efendi, J., Elbanna, A., Dubernet, P., and Abidin, H. Z. (2022). ***Supershear shock front contribution to the tsunami from the 2018 mw 7.5 palu, indonesia earthquake***. Geophys. J. Int., DOI: 10.1093/gji/ggac162.

Jara, J., Bruhat, L., Thomas, M. Y., Antoine, S., Okubo, K., Klinger, Y., Jolivet, R., and **Bhat,** H. S. (2021). ***Signature of transition to supershear rupture speed in coseismic off-fault damage zone***. Proc. R. Soc. A., DOI: 10.1098/rspa.2021.0364.

Elbanna, A., Abdelmeguid, M., Ma, X., Amlani, F., **Bhat,** H. S., Synolakis, C., and Rosakis, A. J. (2021). ***Anatomy of strike slip fault tsunami genesis***. Proc. Natl. Acad. Sci. USA DOI: 10.1073/pnas.2025632118.

**Bhat,** H. S. (2021). ***Supershear earthquakes***. PhD thesis, Habilitation à Diriger des Recherches, Ecole Normale Supérieure.

Jeandet-Ribes, L., Cubas, N., **Bhat,** H. S., and Steer, P. (2020). ***Response of a single fault to transient normal stress change, and implications of large erosional events on the seismic cycle***. Geophys. Res. Lett., 47. DOI: 10.1029/2020gl087631.

Jolivet, R., Simons, M., Duputel, Z., Olive, J.-A., **Bhat,** H. S., and Bletery, Q. (2020). ***Interseismic loading of subduction megathrust drives long-term uplift in northern chile***. Geophys. Res. Lett., DOI: 10.1029/2019gl085377.

Okubo, K., Rougier, E., Lei, Z., and **Bhat,** H. S. (2020). ***Modeling earthquakes with off-fault damage using the combined finite discrete element method***. J. Comp. Part. Mech DOI: 10.1007/s40571-020-00335-4.

Okubo, K., **Bhat,** H. S., Rougier, E., Marty, S., Schubnel, A., Lei, Z., Knight, E. E., and Klinger, Y. (2019). ***Dynamics, radiation and overall energy budget of earthquake rupture with coseismic off-fault damage***. J. Geophys. Res., 124. DOI: 10. 1029/2019jb017304.

Marty, S., Passelègue, F. X., Aubry, J., Schubnel, A., **Bhat,** H. S., and Madariaga, R. (2019). ***Origin of high-frequency radiation during laboratory earthquakes***. Geophys. Res. Lett., 46. DOI: 10.1029/2018gl080519.

Aubry, J., Passelègue, F. X., Deldicque, D., Girault, F., Marty, S., Lahfid, A., **Bhat,** H. S., Escartin, J., and Schubnel, A. (2018). ***Frictional heating processes and energy budget during laboratory earthquakes***. Geophys. Res. Lett., 45. DOI: 10.1029/2018gl079263.

Klinger, Y., Okubo, K., Vallage, A., Champenois, J., Delorme, A., Rougier, E., Lei, Z., Knight, E. E., Munjiza, A., Satriano, C., ***et al.*** (2018). *Earthquake damage patterns resolve complex rupture processes*. Geophys. Res. Lett. DOI: 10.1029/2018gl078842.

Cruz-Atienza, V. M., Villafuerte, C. D., and **Bhat,** H. S. (2018). ***Rapid tremor migration and pore-pressure waves in subduction zones***. Nat. Commun., DOI: 10.1038/s41467-018-05150-3.

Thomas, M. Y., and **Bhat,** H. S. (2018). ***Dynamic evolution of off-fault medium during an earthquake: A micromechanics based model***. Geophys. J. Int., DOI: 10.1093/gji/ggy129.

Romanet, P., **Bhat,** H. S., Jolivet, R., and Madariaga, R. (2018). ***Fast and slow earthquakes emerge due to fault geometrical complexity***. Geophys. Res. Lett DOI: 10.1029/2018gl077579.

Gabuchian, V., Rosakis, A. J., **Bhat,** H. S., Madariaga, R., and Kanamori, H. (2017). ***Experimental evidence that thrust earthquake ruptures might open faults***. Nature, 545. DOI: 10.1038/nature22045.

Thomas, M. Y., **Bhat,** H. S., and Klinger, Y. (2017). ***Effect of brittle off-fault damage on earthquake rupture dynamics***. in *Fault zone dynamic processes : Evolution of fault properties during seismic rupture, geophysical monograph 227*, eds. Thomas, M. Y., **Bhat,** H. S., and Mitchell, T. M., eds., pp. 255–280. American Geophysical Union (AGU). DOI: 10.1002/9781119156895.ch14.

Passelègue, F. X., Latour, S., Schubnel, A., Nielsen, S., **Bhat,** H. S., and Madariaga, R. (2017). ***Precursory processes during laboratory earthquakes***. in *Fault zone dynamic processes : Evolution of fault properties during seismic rupture, geophysical monograph 227*, eds. Thomas, M. Y., **Bhat,** H. S., and Mitchell, T. M., eds., pp. 229–242. American Geophysical Union (AGU). DOI: 10.1002/9781119156895.ch12.

Perol, T., and **Bhat,** H. S. (2016). ***Micromechanics based permeability evolution in brittle materials at high strain rates***. Pure Appl. Geophys., 1–12. DOI: 10.1007/s00024-016-1354-4.

Passelègue, F. X., Schubnel, A., Nielsen, S., **Bhat,** H. S., Deldicque, D., and Madariaga, R. (2016). ***Dynamic rupture processes inferred from laboratory microearthquakes***. J. Geophys. Res., 121. DOI: 10.1002/2015jb012694.

Mello, M., **Bhat,** H. S., and Rosakis, A. J. (2016). ***Spatiotemporal properties of sub-rayleigh and supershear rupture velocity fields : Theory and experiments***. J. Mech. Phys. Solids, DOI: 10.1016/j.jmps.2016.02.031.

Vallage, A., Klinger, Y., Grandin, R., **Bhat,** H. S., and Pierrot-Deseilligny, M. (2015). ***Inelastic surface deformation during the 2013 mw 7.7 balochistan, pakistan, earthquake***. Geology, DOI: 10.1130/g37290.1.

Frank, W. B., Shapiro, N. M., Husker, A. L., Kostoglodov, V., **Bhat,** H. S., and Campillo, M. (2015). ***Along-fault pore-pressure evolution during a slow-slip event in guerrero, mexico***. Earth Planet. Sc. Lett., DOI: 10.1016/j.epsl.2014.12.051.

Siriki, H., **Bhat,** H. S., Lu, X., and Krishnan, S. (2015). ***A laboratory earthquake-based stochastic seismic source generation algorithm for strike-slip faults***. Bull. Seism. Soc. Am., DOI: 10.1785/0120140110.

Mello, M., **Bhat,** H. S., Rosakis, A. J., and Kanamori, H. (2014). ***Reproducing the supershear portion of the 2002 denali earthquake rupture in laboratory***. Earth Planet. Sc. Lett., DOI: 10.1016/j.epsl.2013.11.030.

Passelègue, F. X., Schubnel, A., Nielsen, S., **Bhat,** H. S., and Madariaga, R. (2013). ***From sub-rayleigh to supershear ruptures during stick-slip experiments on crustal rocks***. Science, DOI: 10.1126/science.1235637.

**Bhat,** H. S., Rosakis, A. J., and Sammis, C. G. (2012). ***A micromechanics based constitutive model for brittle failure at high strain rates***. J. Appl. Mech., 79. DOI: 10.1115/1.4005897.

**Bhat,** H. S., Sammis, C. G., and Rosakis, A. J. (2011). ***The micromechanics of westerley granite at large compressive loads***. Pure Appl. Geophys., DOI: 10.1007/s00024-011-0271-9.

**Bhat,** H. S., Biegel, R. L., Rosakis, A. J., and Sammis, C. G. (2010). ***The effect of asymmetric damage on dynamic shear rupture propagation II: With mismatch in bulk elasticity***. Tectonophysics, DOI: 10.1016/j.tecto.2010.03.016.

Biegel, R. L., **Bhat,** H. S., Sammis, C. G., and Rosakis, A. J. (2010). ***The effect of asymmetric damage on dynamic shear rupture propagation i: No mismatch in bulk elasticity***. Tectonophysics, DOI: 10.1016/j.tecto.2010.03.020.

Mello, M., **Bhat,** H. S., Rosakis, A. J., and Kanamori, H. (2010). ***Identifying the unique ground motion signatures of supershear earthquakes: Theory and experiments***. Tectonophysics, DOI: 10.1016/j.tecto.2010.07.003.

Templeton, E. L., **Bhat,** H. S., Dmowska, R., and Rice, J. R. (2010). ***Dynamic rupture through a branched fault configuration at yucca mountain and resulting ground motions***. Bull. Seism. Soc. Am., DOI: 10.1785/012009012110.1785/0120090121.

Harris, R. A., Barall, M., Archuleta, R. J., Dunham, E. M., Aagaard, B., Ampuero, J.-P., **Bhat,** H. S., Cruz-Atienza, V., Dalguer, L., Dawson, P., ***et al.*** (2009). *The SCEC/USGS dynamic earthquake rupture code verification exercise*. Seismol. Res. Lett., 80. DOI: 10.1785/gssrl.80.1.119.

Sammis, C. G., Rosakis, A. J., and **Bhat,** H. S. (2009). ***Effects of off-fault damage on earthquake rupture propagation: Experimental studies***. Pure Appl. Geophys., 166. DOI: 10.1007/s00024-009-0512-3.

Templeton, E. L., Baudet, A., **Bhat,** H. S., Dmowska, R., Rice, J. R., Rosakis, A. J., and Rousseau, C. E. (2009). ***Finite element simulations of dynamic shear rupture experiments and dynamic path selection along kinked and branched faults***. J. Geophys. Res., B08304. DOI: 10.1029/2008jb006174.

Dunham, E. M., and **Bhat,** H. S. (2008). ***Attenuation of radiated ground motion and stresses from three-dimensional supershear ruptures***. J. Geophys. Res., 113. DOI: 10.1029/2007jb005182.

**Bhat,** H. S., Dmowska, R., King, G. C. P., Klinger, Y., and Rice, J. R. (2007b). ***Off-fault damage patterns due to supershear ruptures with application to the 2001 $M_w$ 8.1 kokoxili (kunlun) tibet earthquake***. J. Geophys. Res., B06301. DOI: 10.1029/2006jb004425.

**Bhat,** H. S., Olives, M., Dmowska, R., and Rice, J. R. (2007a). ***Role of fault branches in earthquake rupture dynamics***. J. Geophys. Res., B11309. DOI: 10.1029/2007jb005027.

**Bhat,** H. S. (2007). ***Role of geometric complexities and off-fault damage in dynamic rupture propagation***. PhD thesis, Harvard University.

Fliss, S., **Bhat,** H. S., Dmowska, R., and Rice, J. R. (2005). ***Fault branching and rupture directivity***. J. Geophys. Res., B06312. DOI: 10.1029/2004jb003368.

**Bhat,** H. S., Dmowska, R., Rice, J. R., and Kame, N. (2004). ***Dynamic slip transfer from the denali to totschunda faults, alaska: Testing theory for fault branching***. Bull. Seism. Soc. Am., DOI: 10.1785/0120040601.